

Question #1.

What is the optimal value of alpha for ridge and lasso regression? What will be the changes in the model if you choose double the value of alpha for both ridge and lasso? What will be the most important predictor variables after the change is implemented?

[Ans]

Ridge regression: Below table shows the top predictors and its coefficients when **alpha at its optimal value 10** in case of Ridge regression.

Top predictors: MSZoning_RL, GrLivArea, OverallQual, MSZoning_FV and MSZoning_RM.

	Features	rfe_support	rfe_ranking	Coefficient
11	MSZoning_RL	True	1	0.0877
5	GrLivArea	True	1	0.0791
1	OverallQual	True	1	0.0684
9	MSZoning_FV	True	1	0.0594
12	MSZoning_RM	True	1	0.0580
4	TotalBsmtSF	True	1	0.0462
2	OverallCond	True	1	0.0450
14	Foundation_PConc	True	1	0.0434
7	GarageCars	True	1	0.0355
3	BsmtFinSF1	True	1	0.0328

Ridge regression: Below table shows the top predictors and its coefficients when **alpha optimal value is doubled i.e. 20** in case of Ridge regression.

Top predictors: GrLivArea, OverallQual, MSZoning_RL, MSZoning_FV and TotalBsmtSF.

	Features	rfe_support	rfe_ranking	Coefficient
5	GrLivArea	True	1	0.0710
1	OverallQual	True	1	0.0686
11	MSZoning_RL	True	1	0.0654
9	MSZoning_FV	True	1	0.0478
4	TotalBsmtSF	True	1	0.0455
2	OverallCond	True	1	0.0449
14	Foundation_PConc	True	1	0.0407
12	MSZoning_RM	True	1	0.0380
7	GarageCars	True	1	0.0351
3	BsmtFinSF1	True	1	0.0328

Ridge Observation:

- We can notice the predictors coefficients getting decreased as we tune alpha values to double.

- Which shows as we tune (increase) alpha, few predictors coefficients are shrinking towards zero, hence model tends to be more generalized.

Lasso Regression: Below table shows the top predictors and its coefficients when **alpha at its optimal value 0.0004** in case of Ridge regression.

Top predictors: MSZoning_RL, GrLivArea, MSZoning_RM, MSZoning_FV and OverallQual.

	Features	rfe_support	rfe_ranking	Coefficient
11	MSZoning_RL	True	1	0.109840
5	GrLivArea	True	1	0.107541
12	MSZoning_RM	True	1	0.076511
9	MSZoning_FV	True	1	0.070579
1	OverallQual	True	1	0.069655
4	TotalBsmtSF	True	1	0.046112
2	OverallCond	True	1	0.044889
14	Foundation_PConc	True	1	0.043031
7	GarageCars	True	1	0.036338
3	BsmtFinSF1	True	1	0.033366

Lasso Regression: Here table shows the top predictors and its coefficients when **alpha optimal value is doubled i.e. 0.0008** in case of Ridge regression.

Top predictors: GrLivArea, MSZoning_RL, OverallQual, MSZoning_FV and MSZoning_RM.

	Features	rfe_support	rfe_ranking	Coefficient
5	GrLivArea	True	1	0.106555
11	MSZoning_RL	True	1	0.081213
1	OverallQual	True	1	0.072660
9	MSZoning_FV	True	1	0.055625
12	MSZoning_RM	True	1	0.049598
4	TotalBsmtSF	True	1	0.045558
2	OverallCond	True	1	0.045243
14	Foundation_PConc	True	1	0.039326
7	GarageCars	True	1	0.036137
3	BsmtFinSF1	True	1	0.033912

Lasso Observation:

- In Lasso as well, we can notice the predictors coefficients getting decreased as we tune alpha values to double.
- However, we can also notice that Lasso shrinks the few predictors coefficient to zero thereby removing these variables from the model and hence model tends towards generalization.
- Which shows lasso along with regularization also does feature selection.

Question #2.

You have determined the optimal value of lambda for ridge and lasso regression during the assignment. Now, which one will you choose to apply and why?

[Ans]

Below table shows the performance of the Lasso and Ridge regression on the given dataset for housing prediction.

Evaluation Metrics	Lasso Regression	Ridge Regression
R2score – train	0.9187	0.9182
R2 score – test	0.9025	0.9012
Mean Squared Error - Train data	0.01143	0.01149
Mean Squared Error – Test Data	0.01355	0.01374

As per above metrics its evident that R2 score for Lasso is more compared to the Ridge. As well MSE (Mean squared error) for Lasso is less compared to Ridge.

I will be considering the Lasso regression in this case for my model not only it provides better metrics compared to Ridge, it also helps in feature selection.

Question #3.

After building the model, you realised that the five most important predictor variables in the lasso model are not available in the incoming data. You will now have to create another model excluding the five most important predictor variables. Which are the five most important predictor variables now?

[Ans]

Lasso Regression: The five most important predictor variable after removing previous 5 most predictors from the first model.

Top 5 Predictors are: 2ndFlrSF, 1stFlrSF, TotalBsmtSF, OverallCond and Foundation_PConc.

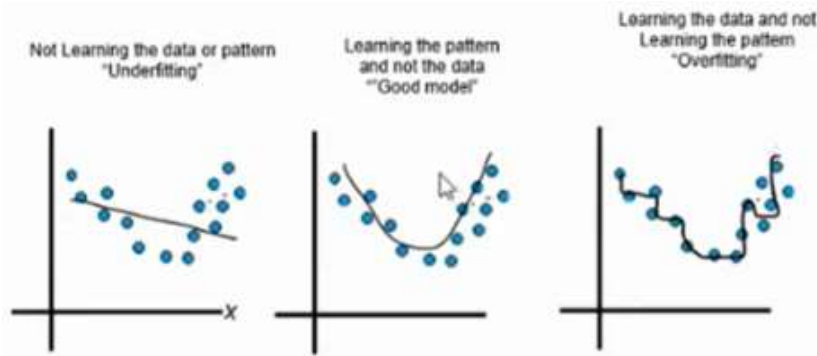
	Features	rfe_support	rfe_ranking	Coefficient
5	2ndFlrSF	True	1	0.096213
4	1stFlrSF	True	1	0.085884
3	TotalBsmtSF	True	1	0.056528
1	OverallCond	True	1	0.056401
14	Foundation_PConc	True	1	0.042647

Question #4.

How can you make sure that a model is robust and generalisable? What are the implications of the same for the accuracy of the model and why?

[Ans]

The general nature of a predictive model is that it does not account for model complexity. It only tries to maximize log-likelihood or reduce cost function, although if it may result in complex coefficients which result in **overfitting**.



Somehow, we try restrain model from becoming rigid. If our model is either “Underfitted” or “Overfitted” then such models are not stable (robust), our aim is to generate models which are stable and generalizable.

To make models robust and generalizable we use regularization.

- **Regularization** can be used to avoid overfitting by creating an optimally complex model. So that it performs well on both test and train data.
- Regularization adds a penalty on the different parameters of the model to reduce the freedom of the model.
- Hence, the model will be less likely to fit the noise of the training data and will improve the generalization abilities of the model.
- In regularized logistic regression the objective function has two parts – the cost function (Log-Likelihood) and regularization term.

$$\hat{\beta} = \min_{\beta} -LL(\beta; y, X) + \lambda R(\beta)$$

- The regularization term has two parts 1> Lambda (hyper parameter) 2> Regularization term.
- By tuning these regularization term and lambda we can **create generalizable and robust models**.

Based on the Regularizer (penalty) we use, there are two methods of regularization:

1. **Ridge Regression:** Here, an addition term of “sum of the squares of the coefficients is added to the cost function” along with the error term.

Ridge Regression

$$\left[\begin{array}{c} \text{Min} \\ \beta \end{array} \right] \left[\begin{array}{c} -LL(\beta; y, X) \\ \text{Negative of Log-Likelihood} \end{array} \right] + \lambda \sum_{i=1}^k \beta_i^2$$

Regularization term
or Regularizer or Penalty

Sum of the squares
of the coefficients

Hyper Parameters

2. **Lasso regression:** Here, a regularization term of “sum of absolute value of the coefficients” is added.

Lasso Regression

$$\left[\begin{array}{c} \text{Min} \\ \beta \end{array} \right] \left[\begin{array}{c} -LL(\beta; y, X) \\ \text{Negative of Log-Likelihood} \end{array} \right] + \lambda \sum |\beta_i|$$

Sum of the absolute values

We have understood that the main reason of overfitting is due to the random errors and fluctuations which are learned by the model while training. The model learns it so well that **training data model's accuracy becomes very high** and resulting in **overfitting**.

We have observed that sudden change in the feature weights resulting in non-linearity of the model. The easy way is to **penalize the feature weights when they get updated**. As explained above two regularization methods make use of this technique.

Here, λ is a hyper parameter whose value is decided by us. **If λ is high it adds high penalty to error term making the learned hyper plane almost linear**, if λ is close to 0 it has almost no effect on the error term causing no regularization. **So, these regularization methods help us to get optimum coefficients such that model accuracy on Train and Test data will be high and almost nearby.**

Also, in Lasso regularization helps in feature selection technique too as it zero out the respective weights of undesired features, but it is computationally expensive.